

Daniel Zhou

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EDUCATION

University of California, Los Angeles (UCLA)
B.S. Computer Science and Engineering

June 2027
GPA **3.98/4.0**

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Computer Architecture, Machine Learning, Artificial Intelligence

SKILLS

Languages: C++, Python, Java, Kotlin, TypeScript, JavaScript, SQL, MATLAB

Frameworks: React, Next.js, React Native, Jetpack Compose, Node.js, FastAPI, LangGraph

Technologies: AWS, Docker, MCP, Supabase, MongoDB, Git, GitHub, LangSmith, Stripe, MQTT, ESP32

EXPERIENCE

Leechy (Rental Marketplace)

07/2025–Present

Lead Android Engineer

- Architected and built Leechy's production Android marketplace from the ground up as the primary Android engineer, developing a ~135K-line Kotlin/Jetpack Compose app and owning Google Play releases and deployment.
- Engineered end-to-end rental and service workflows spanning multi-state bookings, payments, delivery, handoff verification, condition tracking, messaging, and availability.
- Built geospatial discovery with Mapbox, dynamic clustering, and Google Places; integrated Stripe Payment Sheet, Connect, and Identity for checkout, seller onboarding, payouts, and subscriptions.
- Designed layered MVVM architecture and asynchronous data flows with Kotlin coroutines and Supabase while co-developing backend APIs shared across Android, iOS, and web clients.

Trustworthy AI Lab at UCLA

06/2026–Present

Undergraduate Researcher

- Built the contestant SDK and autonomous-agent runtime for a multi-game competition platform, handling JWT authentication, automatic match and tournament discovery, MCP gameplay, structured actions, retries, and isolated concurrent match execution.
- Designed cross-game MCP and runtime-adaptor abstractions supporting heterogeneous environments and messaging workflows; validated the real contestant execution path across 5+ game/runtime behaviors with ~205 automated tests and deployed end-to-end tests.
- Deployed and debugged production AWS infrastructure using ECS/Fargate, Docker, VPC networking, HTTPS, and WebSockets, including private Pokémon Showdown hosting for fully cloud-based battles.
- Built end-to-end observability for a LangGraph reasoning agent, tracing structured failures across runtime streams and Supabase with correlation IDs, error classification, and LangSmith trace linking.
- Re-architected multi-turn LangGraph state management to reconstruct agent state from durable messages and table metadata, eliminating persistent checkpoints and redundant execution history while preserving cross-turn behavior.

Rocket Project at UCLA

09/2023–06/2026

Controls Lead / Electrical Ground Support Equipment Lead

- Led development and hotfire validation of the first successful closed-loop liquid rocket engine throttling system in Rocket Project's history, integrating embedded C++ PID control, custom actuated valves, pressure feedback, and fault detection/recovery.
- Designed and experimentally validated actuated valve hardware and tuned closed-loop pressure controllers through simulation, HIL, waterflow, and coldflow testing, achieving <0.7 s setpoint settling.
- Developed Python/MATLAB propulsion simulations modeling feed-system flow, chamber pressure, mass flow, thrust, and valve dynamics to determine attainable setpoints and tune controller behavior.
- Designed embedded ground-control and DAQ systems using ESP32/Arduino, custom PCBs, and 16-channel pressure acquisition; developed Rocket Project's first MQTT-based wireless control and telemetry system for remote engine operations.

Creative Labs at UCLA

09/2024–06/2025

Mobile Developer Lead

- Led a 3-developer team building a gamified coding education app with React Native, translating Figma designs into interactive features for problems, rankings, progression, and in-app currency.
- Owned frontend features from design handoff through backend integration, collaborating with designers and developers to integrate REST APIs and debug cross-stack issues.